Future development plans for tourism under the Corona-virus disease

Tourism is a compound industry with a high degree of relevance. It is not only directly related to the transportation industry, catering industry, and tourism service industry, but also closely related to most sectors of the tertiary industry. The consumption expenditure driven by the tourism industry mainly includes three parts: the expenditure of tourists on the purchase of goods and services at the tourist destination, the transportation, accommodation, catering, entertainment, leisure and other services generated during the tourism process, the construction of tourist attractions and tourism Various infrastructure costs of enterprises, etc.

However, the pandemic of the COVID-19 has had varying degrees of impact on all walks of life. Which has a greater negative impact on the tourism industry. The intentional or unintentional measures taken by some countries to prevent the spread of the epidemic have severely inhibited the development of tourism. The depression of the tourism industry will inevitably affect the labor desire of tourists and the income of tourism enterprises and government departments, which in turn will lead to economic stagnation or regression. In the context of the COVID-19, some countries have also taken the initiative to explore new tourism models, but believe that a general consensus has been reached. For a country like China with a huge population and multiple holidays of varying duration, how to weigh the interests of all parties and then propose a scientific tourism development plan in the context of the epidemic appears to be very meaningful.

Attachment 1 gives the names and latitude and longitude information of some scenic spots in Beijing. Please solve the following five problems in combination with the related theories of mathematical modeling:

Question 1: Refer to the data in Annex 1. Please collect the basic data of scenic spots in various provinces across the country. It is especially necessary to collect the current evaluation level of scenic spots and the maximum reception capacity per day, and visually display the distribution characteristics of these scenic spots.
Question 2: Can you use the collected attractions and reception capabilities of these scenic spots to quantitatively evaluate and classify them reasonably?
Question 3: Under the background of the epidemic situation, the current limit of each scenic spot has become a new model. Can you provide a reasonable current limit model for the scenic spot according to different epidemic levels, and give a specific quantitative analysis result using a certain scenic spot as an example?
Question 4: Can you use mathematical modeling methods to propose a quantitative model that also considers epidemic prevention and control, tourism income, and tourist experience, and conduct simulation analysis of the expected effects.
Question 5: Can you provide government management with a differentiated management plan for different scenic spots during the epidemic period of no more than 2 pages?

Your submission should consist of:

- One-page Summary Sheet
- Table of Contents
- One -to Two -page Article
- Your solution of no more than 20 pages, for a maximum of 24 pages with your summary, table of contents, and article

Note: Reference List and any appendices do not count toward the page limit and should appear after your complete solution. You should not make use of unauthorized images and materials whose use is restricted by copyright laws. Ensure you cite the sources for your ideas and the materials used in your report.